TASK 2: OWA

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1 Description

The aim of this second task is to implement two basic methods of aggregation, the Weighted Sum Model (WSM), which is equivalent to the weighted average, and the Ordered Weighted Average (OWA) operator. The decision problem involves selecting the best restaurant based on several criteria. The selected restaurants are located in Tarragona, and six criteria are used to evaluate each restaurant.

Restaurant	Food	Personnel	Atmosphere	Category	Location	Terrace
Terrat	0.67	0.51	0.55	0.72	0.80	0.90
RacoAbat	0.60	0.45	0.50	0.67	0.75	0.85
Dominos	0.40	0.35	0.45	0.60	0.70	0.80
Ancora	0.65	0.55	0.60	0.70	0.77	0.85
Frida	0.50	0.40	0.55	0.68	0.78	0.88
Barhaus	0.75	0.65	0.70	0.80	0.85	0.95

Table 1: Evaluation of Restaurants based on six criteria.

In this evaluation, the most important criteria are **Food** and **Personnel**, each receiving a weight of 0. The remaining criteria **Atmosphere**, **Category**, **Location**, and **Terrace** are each assigned a weight of 0.1, indicating they are considered less and important than **Food** and **Personnel** but still relevant for the overall assessment and equivalent between them.

Criteria	Weight
Food	0.3
Personnel	0.3
Atmosphere	0.1
Category	0.1
Location	0.1
Terrace	0.1

Table 2: Weights assigned to each criterion in the evaluation process.

The initial assessment of the restaurants can be made visually by analyzing the area of the radar plots, which give a clear indication of each restaurant's performance across the criteria. The plot below shows the radar plot for the restaurants.

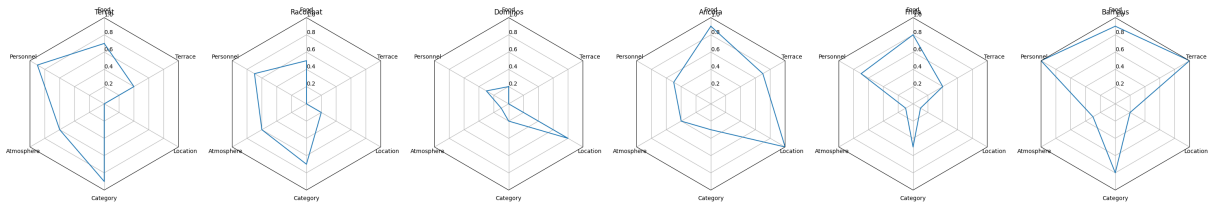


Figure 1: Radar plot of the evaluated restaurants. The larger the area, the better the performance.

At first sight, it is evident that **Terrat**, **Ancora**, and **Barhaus** stand out with larger areas, suggesting that these are the top-performing restaurants based on the criteria used in the evaluation.

2 Task 1: Weighted Average Analysis

As shown in Figure 2, which show the WSM results, we can observe that Barhaus has the highest utility score, followed by Terrat and Ancora, which have almost identical values. This outcome aligns with what we visually inferred from the radar plots, where these three alternatives were the ones with highest area. The radar plots provided an initial indication of the general ranking, with Barhaus standing out clearly as the top choice, and Terrat and Ancora closely trailing behind,

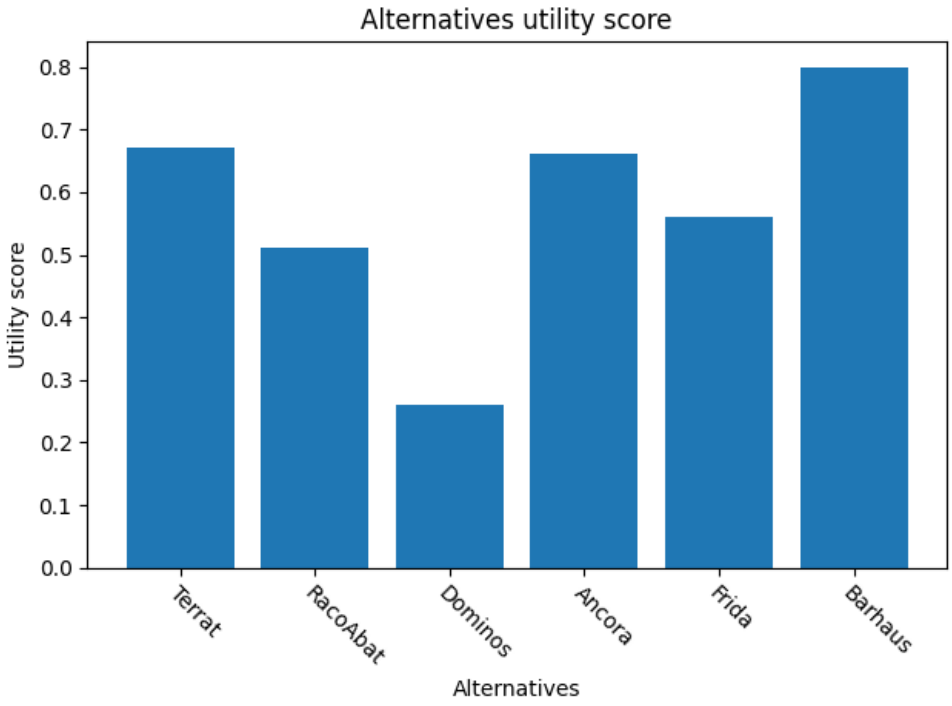


Figure 2: Utilities of the Alternatives using the WSM Method

3 Task 2: Adding new alternatives

In this task, two new restaurants are added. Which are **Restaurante BARMUT** and **McDonalds**. On one hand, **McDonalds** is characterized by having the best evaluation in the **Location** criterion, with a score of 0.9, but it is rated poorly in all other criteria, receiving scores ranging from 0.2 to 0.4. On the other hand, **Restaurante BARMUT** stands out for its very good performance in **Personnel** (service), with a score of 0.9, but it scores poorly in all the other criteria, similar to **McDonalds**.

Figure 3 shows the radar plot of the newly added alternatives, which can be observable that have noticeably low area, which indicate that a priori are not good alternatives.

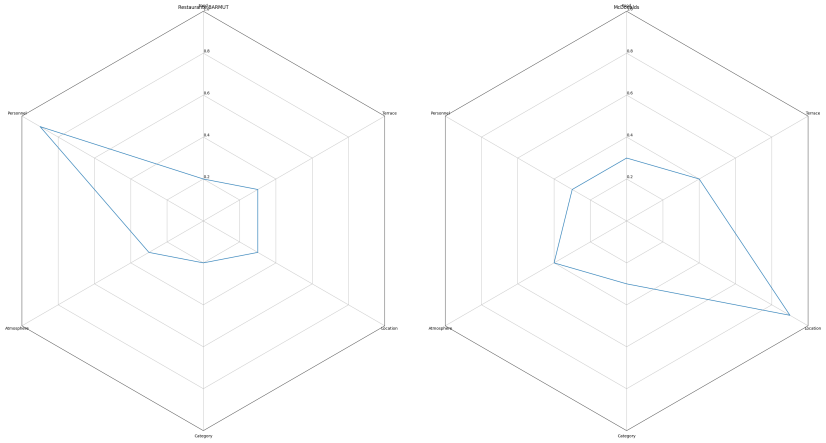


Figure 3: Radar plot comparing the new alternatives across all criteria.

Finally, Figure 4 provides the bar plot of all the alternatives afterward. The first 6 columns are the initial alternatives, which are not modified compared to those in task 1. However, the last two columns are the newly added alternatives that show their poor averaged utility score that performs just better than the worse alternative, which is **Dominos**.

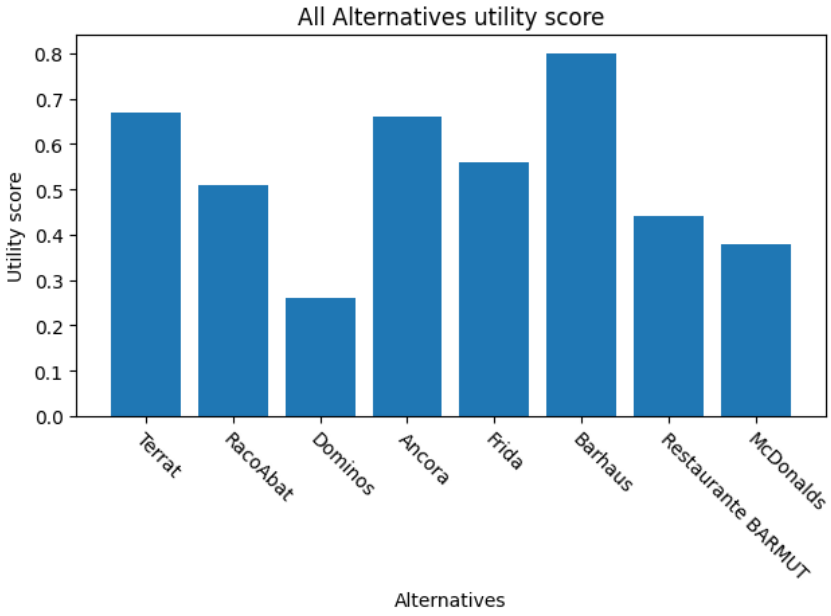


Figure 4: Utility values for the alternatives.

4 Task 3: Arithmetic Non-weighted Average

In task 3 the same procedure as in task 1 is applied, but after modifying the weights to make all criteria have the same weight.

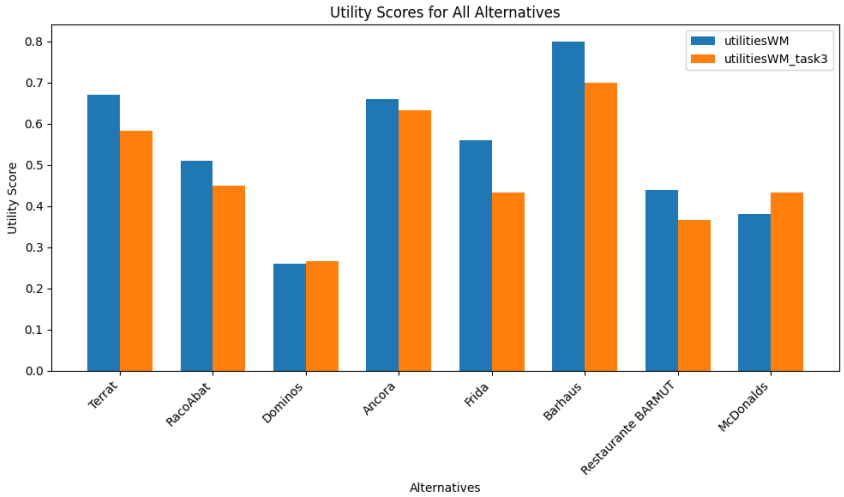


Figure 5: Bar plot comparing utilities with the Weighted Average (utilitiesWM) and the Non-weighted Average (utilitiesWM_task3).

The results from Figure 5 show slight differences in the utility scores. In blue, all the alternatives have weighted average and in orange, using non-weighted average. In the figure, all options except **McDonald's** and **Dominos** performed worse in this third task. This decline is likely due to the **Location** criterion being given more weight. Initially, it had a weight of 0.1, but in this task, all criteria were weighted equally, making **Location** account for approximately 0.1667 of the total. This new weighting favors the restaurants that had poorer scores in the location criterion. Despite this, **Barhaus** remains the best restaurant, followed by **Terrat** and **Ancora**. However, **Ancora** now scores better than **Terrat**, as a result of the new weight distribution across the criteria.

5 Task 4: OWA operator

This fourth task invovles practicing with the Ordered Weighted Averaging (OWA) operator, which is a type of aggregation function that works a by assigning different weights to ordered values rather than to specific criteria. The steps in the OWA operator usually involve first sorting the input values in descending order and then applies predefined weights to compute a weighted sum.

5.1 OWA Example

1. **Student's Marks:** A student obtains the following marks on four assignments in the same subject: (9, 3, 5, 7).
2. **Ordered Marks:** Arranging them in decreasing order: (9, 7, 5, 3).
3. **Arithmetic Average:** 6

Applying the OWA Operator:

1. **Example 1: Emphasizing Low Marks**
   Using the weight vector $w = (0.2, 0.2, 0.2, 0.4)$:
   $$0.2 \times 9 + 0.2 \times 7 + 0.2 \times 5 + 0.4 \times 3 = 5.4$$
   Since the lowest mark (3) has the highest weight, the final score is reduced.
2. **Example 2: Prioritizing Medium Values**
   Using the weight vector $w = (0, 0.5, 0.5, 0)$:
   $$0 \times 9 + 0.5 \times 7 + 0.5 \times 5 + 0 \times 3 = 6$$
   Here, the highest (9) and lowest (3) marks are ignored, while the middle values (7 and 5) are weighted equally, leading to a balanced result.

This example shows how the choice of weights influences the final score. Giving more weight to lower marks lowers the overall result, while focusing on middle values leads to a more balanced outcome.

5.2 OWA Task 4 Analysis

In this section we start with the same alternatives and criteria as described in Table 1 and the weights as:

$$\text{Weights} = [0.1, 0.3, 0.6, 0.0, 0.0, 0.0]$$

These weights show us that just the top 3 performing criteria for each restaurant are considered and the other 3 least performing criteria are not considered. Let's see the example for the restaurant Terrat which has the following utility values:

$$\text{Utility Values} = [0.7, 0.9, 0.6, 0.9, 0.0, 0.4]$$

To compute the OWA, we follow these steps:

1. **Sort the values in descending order:**

$$\text{Sorted Values} = [0.9, 0.9, 0.7, 0.6, 0.4, 0.0]$$

2. **Compute the weighted sum:** The OWA is the sum of the sorted values multiplied by their corresponding weights:

$$\text{OWA} = 0.9 \times 0.1 + 0.9 \times 0.3 + 0.7 \times 0.6 + 0.6 \times 0.0 + 0.4 \times 0.0 + 0.0 \times 0.0$$

3. **Final computation:**

$$\text{OWA} = 0.09 + 0.27 + 0.42 + 0.0 + 0.0 + 0.0 = 0.78$$

Thus, the OWA for the Terrat restaurant is 0.78.

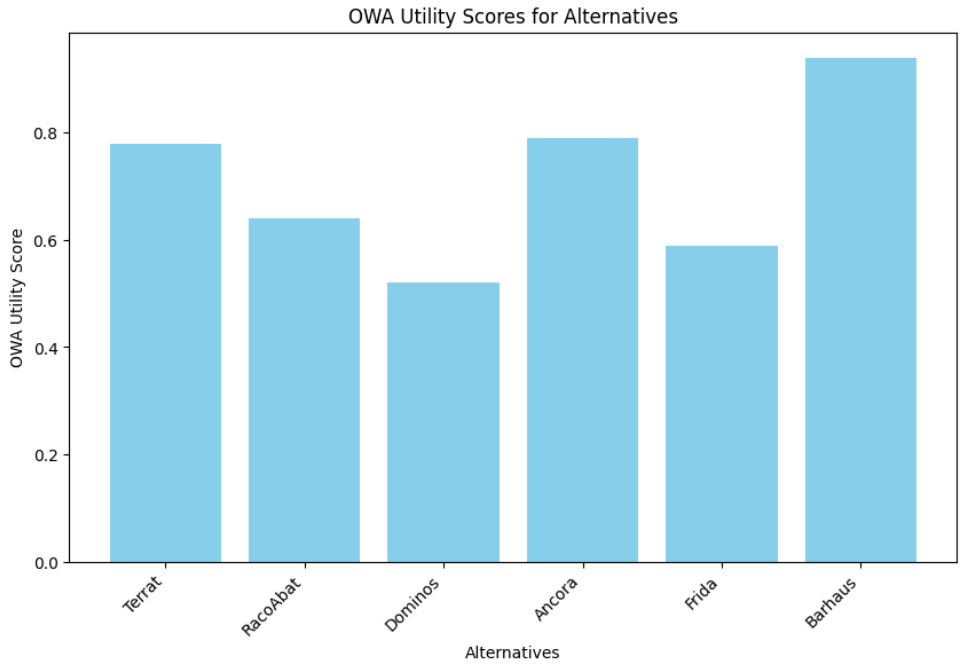


Figure 6: Utility bar plot with OWA.

The resulting utilities after they are calculated for all restaurants are shown in Figure 6. A quick analysis on this figure shows us that with this weight assignment, the behavior is the same as the one obtained in Task 1 as shown on Figure 2. Next, we define a Disjunctive (OR) and a Conjunctive (AND) policy. The Disjunctive policy gives higher weights to the best values while assigning zero weight to the two worst values. The weights are:

$$\text{Weights (Disjunctive)} = [0.3, 0.3, 0.2, 0.2, 0.0, 0.0]$$

The Conjunctive policy assigns weights evenly, but with slightly more weight on the better values. The weights are:

$$\text{Weights (Conjunctive)} = [0.2, 0.2, 0.2, 0.2, 0.1, 0.1]$$

Next, Figure 7 shows the comparison between the Disjunctive and Conjunctive Policy.

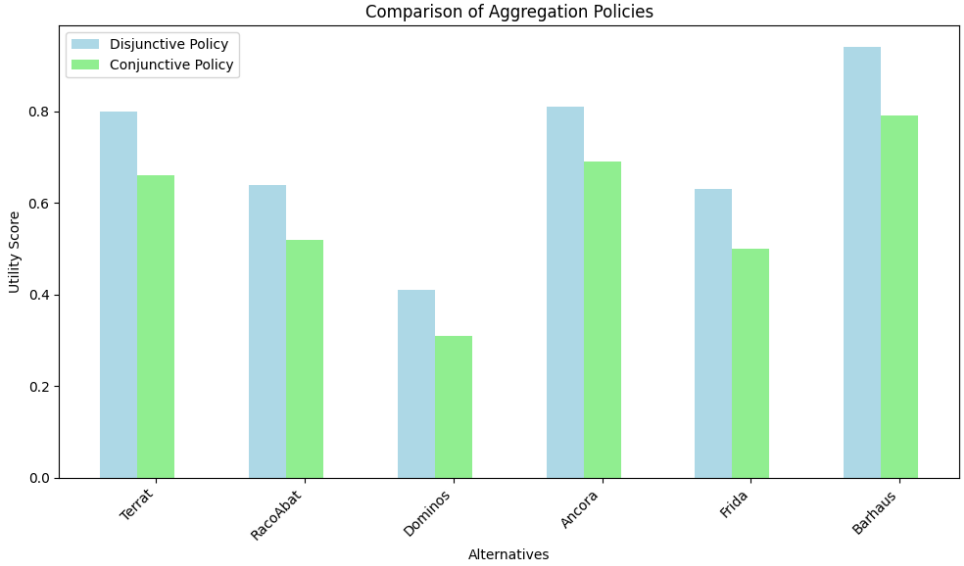


Figure 7: Comparison Conjunctive vs Disjunctive policy.

The results show that the Disjunctive Policy has better utilities for all alternatives while the Conjunctive has worse ones. This makes sense as the Disjunctive policy removes the worse criteria for all alternatives, making them score better. While the Conjunctive one considers all good and bad criteria, obtaining a more averaged result.

6 Task 5: Aggregation policies, descriptors of the OWA weights

In this section we will study the different OWA descriptors which are:

1. **Orness:**

$$\text{Orness or Attitudinal Character} = \sum_{j=1}^{n} \frac{n+1-2j}{n-1} \cdot w_j$$

   Orness is a measure between 0 and 1:

   - The higher the Orness, the more *Disjunctive* (focuses on the best values).
   - The lower the Orness, the more *ANDness* (Conjunctive, treating all criteria more equally).

2. **Balance:**

$$\text{Balance} = \sum_{j=1}^{n} \frac{n+1-2j}{n-1} \cdot w_j$$

   Balance is between -1 and 1:

   - -1 represents the minimum operator (no preference for any criterion).
   - 1 represents the maximum operator (one criterion is dominant).
   - 0 represents arithmetic averaging, where all criteria are treated equally.

3. **Entropy:**

$$\text{Entropy} = -\sum_{j=1}^{n} w_j \log(w_j)$$

   Entropy is a measure of uncertainty:

- High entropy occurs when $w_j = \frac{1}{n}$ for all j, meaning the distribution is uniform.
- Low entropy occurs when $w_q = 1$ and all other $w_j = 0$ for $j \neq q$, meaning all the weight is focused on a single criterion.

4. **Divergence:**

$$\text{Divergence} = \sum_{j=1}^{n} w_j \cdot \left(\frac{n-j}{n-1}\right) \cdot (-\alpha(W))^2$$

Divergence is a measure of how the weights deviate from a uniform distribution:

- Divergence is 0 for optimistic and pessimistic cases when $w_j = 1$, i.e., all weight is concentrated on one criterion (complete certainty).

6.1 OWA Descriptors Analysis

The different OWA weights to be analysed are shown in Figure 8, where the values are already computed.

	OWA Weight Set	Weights	Balance	Divergence	Entropy	Orness
0	OWA Example 1	[0.0, 0.3, 0.4, 0.3, 0.0, 0.0]	0.20	0.0240	1.088900	0.60
1	OWA Example 2	[0.0, 0.0, 0.0, 0.2, 0.3, 0.5]	-0.72	0.0244	1.029653	0.14
2	OWA Example 3	[1.0, 0.0, 0.0, 0.0, 0.0, 0.0]	1.00	0.0000	-0.000000	1.00
3	OWA Example 4	[0.1, 0.1, 0.3, 0.3, 0.1, 0.1]	0.00	0.0740	1.643418	0.50
4	OWA Example 5	[0.1, 0.3, 0.6, 0.0, 0.0, 0.0]	0.40	0.0180	0.897946	0.70

Figure 8: Descriptor values for the OWA weight examples

6.2 Balance Analysis:

As previously explained, the balance ranges between -1 and 1. A value of -1 means a highly pessimistic (minimum operator), 1 means a highly optimistic (maximum operator), and 0 represents an arithmetic average of values. Therefore, positive values indicate a tendency toward an optimistic decision, favoring higher values in the aggregation.

From Figure 8 we can see that the **OWA Examples 1, 3 and 5** obtain a positive balance, while **OWA example 2** obtains a negative balance and **OWA example 4** a neutral one. **OWA example 3** is a clear case of a maximum operator, where just one criterion obtains all the weight, a clear example of a disjunctive case. On the other hand, **OWA example 2** as obtains a negative result, we can see that it is a weighting that has preference over lower values rather than higher values. Thus, the neutral weighting showed as **OWA example 4** means that there is no preference between high and low values.

6.3 Orness Analysis:

Orness is takes values between 0 and 1. 0 means the policy is highly Conjunctive (i.e., requiring most criteria to be fulfilled), and 1 means the policy is highly Disjunctive (i.e., only one criterion needs to be fulfilled).

In this second metric, we can infer similar conclusions as those obtained with the Balance Analysis. **OWA Examples 1, 3 and 5** obtain a Orness higher than 0.5, **OWA Example 4** obtains a neutral Orness value of 0.5 and **OWA Examples 2** obtains an orness lower than 0.5 indicating its preference over smaller values.

6.4 Divergence Analysis:

Divergence values are positive, with 0 indicating no divergence (perfect uniformity), and larger values indicating higher disparity between values. A low divergence suggests that the distribution of weights among positions is relatively even, while a higher divergence indicates a more polarized distribution, with some positions being highly favored over others.

OWA Example 1, with a very low divergence of 0.0240, suggests a nearly uniform distribution with minimal polarization and only a slight preference for certain values. Similarly, **OWA Example 2**, with a divergence of 0.0244, also exhibits low polarization, indicating a balanced distribution. **OWA Example 3**, however, has a divergence of 0.0000, meaning all weight is concentrated at the highest value, with no spread among the weights. In contrast, **OWA Example 4** shows a higher divergence of 0.0740, reflecting greater polarization, where weights are more evenly spread across the middle range with some small weights at the extremes. Lastly, **OWA Example 5**, with a divergence of 0.0180, represents a moderate spread of weights, slightly favoring higher values due to the influence of the 0.6 weight but without strong skewness toward one end.

6.5 Entropy Analysis:

Entropy measures the spread or "disorder" of the weights, with high entropy indicating a relatively even spread of weights across positions and low entropy indicating a highly concentrated weight distribution. When $W_j = \frac{1}{n}$ for all j, entropy is at its highest, while low entropy occurs when one weight dominates.

In this case the results are similar than those obtained with the divergence analysis. The entropy values in the OWA examples indicate the level of balance in the weight distributions. **OWA Example 1**, with an entropy of 1.0889, suggests a relatively high and balanced spread of weights, where no single weight dominates the others, typical for sets with more evenly distributed weights. Similarly, **OWA Example 2**, with an entropy of 1.0297, also reflects a high but slightly more concentrated distribution compared to **OWA Example 1**. In contrast, **OWA Example 3** has an entropy of -0.0, indicating a perfectly concentrated weight set with no spread, as all weight is placed on the first position, leading to minimal entropy. On the other hand, **OWA Example 4**, with an entropy of 1.6434, represents a very high and balanced distribution of weights, where all weights are more or less equal. Lastly, **OWA Example 5**, with an entropy of 0.8979, indicates a moderate level of entropy, reflecting a relatively even distribution but with a slight concentration toward higher values..